

$$\frac{97}{300} = 0.323, \frac{299}{500} = 0.598.$$

Clearly, each one of 0.39 and 0.7175 lies between 0.333 and 0.75.

So, $\frac{117}{300}$ and $\frac{287}{400}$ lie between $\frac{1}{3}$ and $\frac{3}{4}$.

26. $\frac{1}{5} = 0.2$ and $\frac{1}{6} = 0.166$.

(a) $\frac{4}{25} = 0.16, \frac{9}{50} = 0.18, \frac{17}{100} = 0.17$

(b) $\frac{4}{25} = 0.16, \frac{17}{100} = 0.17, \frac{167}{1000} = 0.167$

(c) $\frac{4}{25} = 0.16, \frac{9}{50} = 0.18, \frac{167}{1000} = 0.167$

(d) $\frac{9}{50} = 0.18, \frac{17}{100} = 0.17, \frac{167}{1000} = 0.167$

27. $\begin{array}{r} 0.3 \\ 3.0 \\ 3.33 \\ 3.03 \\ \hline + 333.0 \\ \hline 345.96 \end{array}$	28. $\begin{array}{r} 636.66 \\ 366.36 \\ + 363.33 \\ \hline 1366.35 \end{array}$	29. $\begin{array}{r} 24.424 \\ 5.656 \\ 1.131 \\ + 0.089 \\ \hline 31.300 \end{array}$
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30. $\begin{array}{r} 555.05 \\ 55.5 \\ 5.55 \\ 5.0 \\ \hline + 0.55 \\ \hline 621.65 \end{array}$	31. $\begin{array}{r} 4.0 \\ 4.44 \\ 0.4 \\ 44.04 \\ \hline + 444.0 \\ \hline 496.88 \end{array}$	32. $\begin{array}{r} 999.99 \\ 999.9 \\ + 99.9 \\ + 99.9 \\ \hline 1109.7 \end{array}$
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33. $\begin{array}{r} 10.0001 \\ + 9.9999 \\ \hline 20.0000 \end{array}$	34. $\begin{array}{r} 48.95 \\ - 32.006 \\ \hline 16.944 \end{array}$
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35. Given expression = $58.621 - (13.829 + 7.302 + 1.214)$
 $= 58.621 - 22.345 = 36.276$.

36. Let $341.42 - 53.74 = x - 62.86$.

Then, $x = (341.42 + 62.86) - 53.74$
 $= 404.28 - 53.74 = 350.54$.

37. Let $534.596 + 61.472 - 496.708 = x + 27.271$

Then, $x = (534.596 + 61.472) - (496.708 + 27.271)$
 $= 596.068 - 523.979 = 72.089$.

38. Given expression = $(11.71 + 1.78) - (0.86 + 9.20)$
 $= 13.49 - 10.06 = 3.43$.

39. Let $726.34 + 888.12 - x = 1001.88$.

Then, $x = (726.34 + 888.12) - 1001.88$
 $= 1614.46 - 1001.88 = 612.58$.

40. Let $832.58 - 242.31 = 779.84 - x$.

Then, $x = (779.84 + 242.31) - 832.58$
 $= 1022.15 - 832.58 = 189.57$.

41. (a) $\frac{9}{4} + 1.75 = 2.25 + 1.75 = 4$.

(b) $\frac{9}{5} + 2.2 = 1.8 + 2.2 = 4$.

(c) $\frac{6}{5} + 2.8 = 1.2 + 2.8 = 4$.

(d) $\frac{3}{2} + 1.5 = 1.5 + 1.5 = 3$.

42. (a) $x^2 + y^2 = (10)^2 + (0.1)^2 = 100 + 0.01 = 100.01$.

(b) $x^2 - y^2 = (10)^2 - (0.1)^2 = 100 - 0.01 = 99.99$.

(c) $x^2 y^2 = (10)^2 \times (0.1)^2 = 100 \times 0.01 = 1$.

(d) $\frac{x^2}{y^2} = \frac{(10)^2}{(0.1)^2} = \frac{100}{0.01} = 10000$.

Clearly, $\frac{x^2}{y^2}$ is the greatest.

43. (a) $(0.09)^2 = 0.0081$

(b) 0.09

(c) $(1 - 0.9)^2 = (0.1)^2 = 0.01$

(d) $1 - (0.9)^2 = 1 - 0.81 = 0.19$

Clearly, $0.0081 < 0.01 < 0.09 < 0.19$.

So, 0.0081 is closest to zero.

44. Required fraction = $\frac{\frac{1}{100} \text{ cm}}{1 \text{ km}} = \frac{\left(\frac{1}{100}\right) \text{ cm}}{(1000 \times 100) \text{ cm}}$
 $= \frac{1}{100 \times 1000 \times 100}$
 $= \frac{1}{1000000} = 0.0000001$.

45. Let $x + 543 + 5.43 = 603.26$.

Then, $x = 603.26 - (543 + 5.43)$
 $= 603.26 - 548.43 = 54.83$.

$\therefore$ Missing digit = 8.

46. $3.14 \times 10^6 = 3.140000 \times 1000000 = 3140000$.

47. $518,000,000 = 5.18 \times 100000000 = 5.18 \times 10^8$.

48. $0.000006723 = \frac{0.000006723 \times 10^6}{10^6}$
 $= \frac{6.723}{10^6} = 6.723 \times 10^{-6}$.

49. $10^k = \frac{0.001125}{1.125} = \frac{1.125}{1125} = \frac{1.125 \times 10^3}{1125 \times 10^3} = \frac{1}{10^3} = 10^{-3}$.

$\therefore k = -3$.

50. Given expression = $\left(\frac{1}{10} \times \frac{1}{100} \times \frac{1}{1000} \times 10^7\right)$
 $= \left(\frac{10^7}{10^6}\right) = 10^{(7-6)} = 10$.

51. $383 \times 38 \times 38 = 553052$.

Number of decimal places = 1.

$\therefore 383 \times 38 \times 3.8 = 55305.2$.

52. $9 \times 7 = 63$.

Sum of decimal places = 5. $\therefore 0.09 \times 0.007 = 0.00063$.

53. Sum of decimal places = 6. $\therefore 0.1 \times 0.01 \times 0.001 = 0.000001$.

54. $3 \times 3 \times 3 \times 3 \times 30 = 2430$.